

Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

Members:

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Date: November 21, 2025

Maximize

$$Z = 1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10}$$

Subject to

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \leq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \leq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \leq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \leq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \leq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \leq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \geq 4.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} = 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \leq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \geq 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} = 0.00$$

$$1.00x_1 + 1.00x_2 + 1.00x_3 + 1.00x_4 + 1.00x_5 + 1.00x_6 + 1.00x_7 + 1.00x_8 + 1.00x_9 + 1.00x_{10} \geq 0.00$$

Simplex Table

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	0	0	0	0	0	0	0	0	0	0	M	M	M	M	M	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Artificial Variables "Canonization"

Variable a_1

$$R_1 \leftarrow R_1 + -4MR_8$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-1M	-1M	-1M	-1M	-1M	-1M	-1M	-1M	-1M	-1M	0	0	0	0	0	0	0	M	0	0	0	M	M	M	M	-4M
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	1	0

Variable a_2

$$R_1 \leftarrow R_1 + -0R_9$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-2M	-2M	-2M	-2M	-2M	-2M	-2M	-2M	-2M	-2M	0	0	0	0	0	0	0	M	0	0	0	0	M	M	M	-4M
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	1	0

Variable a_3

$$R_1 \leftarrow R_1 + -0R_{11}$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-3M	-3M	-3M	-3M	-3M	-3M	-3M	-3M	-3M	-3M	0	0	0	0	0	0	0	M	M	0	0	0	0	M	M	-4M
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Variable a_4

$$R_1 \leftarrow R_1 + -0R_12$$

	Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-4M	-4M	-4M	-4M	-4M	-4M	-4M	-4M	-4M	-4M	-4M	0	0	0	0	0	0	0	M	M	0	0	0	0	0	M	-4M
0	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Variable a_5

$$R_1 \leftarrow R_1 + -0R_13$$

	Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	0	0	0	0	0	0	0	M	M	M	0	0	0	0	0	-4M
0	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Intermediate Tables

Pivoting 1

Most Negative

Column 2 (-5M)

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b	
1	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	0	0	0	0	0	0	0	0	M	M	M	0	0	0	0	0	-4M
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	4	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0	

Fractions

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b	Frac
1	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	0	0	0	0	0	0	0	M	M	M	0	0	0	0	0	-4M	
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4	4.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	0	0	1	0	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0	0.00

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$4.00/1.00 = 4.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

Smallest fraction: 0.00 $\rightarrow$ Pivot: row 13

Pivot

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	0	0	0	0	0	0	0	M	M	M	0	0	0	0	0	-4M
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1

Canonization

$$R_1 3 \leftarrow R_1 3 / 1.00$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	-5M	0	0	0	0	0	0	0	M	M	M	0	0	0	0	0	-4M
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1

$$R_1 \leftarrow R_1 + 5000001.00 R_1 3$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_2 \leftarrow R_2 + -1.00 R_1 3$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_3 \leftarrow R_3 + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_4 \leftarrow R_4 + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_5 \leftarrow R_5 + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_6 \leftarrow R_6 + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_7 \leftarrow R_7 + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	4
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_8 \leftarrow R_8 + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_9 \leftarrow R_9 + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_{10} \leftarrow R_{10} + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_{11} \leftarrow R_{11} + -1.00R_13$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_{12} \leftarrow R_{12} + -1.00R_{13}$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Pivot Result

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Degenerate Base

The variable x_1 is part of the base but has a value of 0. Therefore, this is a degenerate Basic Feasible Solution (BFS).

Pivoting 2

Most Negative

Column 21 (-4M)

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Fractions

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b	Frac
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M	
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4	4.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0	0.00
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0	0.00
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0	-0.00

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$4.00/1.00 = 4.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/1.00 = 0.00$$

$$0.00/-1.00 = -0.00$$

Smallest fraction: 0.00 $\rightarrow$ Pivot: row 12

Pivot

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

Canonization

$$R_12 \leftarrow R_12/1.00$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	-4M	0	0	0	0	5M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_1 \leftarrow R_1 + 4000001.00R_12$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_2 \leftarrow R_2 + -1.00R_12$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b		
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M	
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	-1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	0	-1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0	
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0	

$$R_3 \leftarrow R_3 + -1.00R_12$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_4 \leftarrow R_4 + -1.00R_12$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	1	0	0	0	-1	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_5 \leftarrow R_5 + -1.00R_12$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	0	0	-1	0	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_9 \leftarrow R_9 + -1.00R_{12}$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	0	0	-1	0	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_{10} \leftarrow R_{10} + -1.00R_{12}$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	0	0	-1	0	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	1	0	0	-1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	1	0

$$R_{11} \leftarrow R_{11} + -1.00R_{12}$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	0	0	-1	0	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	1	-1	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	1	0

$$R_13 \leftarrow R_13 + 1.00R_12$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	0	0	-1	0	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	1	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Pivot Result

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	s_1	s_2	s_3	s_4	s_5	s_6	s_7	e_1	e_2	e_3	a_1	a_2	a_3	a_4	a_5	b
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	M	M	0	0	0	0	4M	M	-4M
0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	0	0	-1	0	4
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	1	-1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	-1	0
0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Degenerate Base

The variable x_1 is part of the base but has a value of 0. Therefore, this is a degenerate Basic Feasible Solution (BFS).

Non-Feasible

The artificial variable a_3 is in the base. Therefore, a BFS (basic-feasible-solution) does not exist for this problem.